README

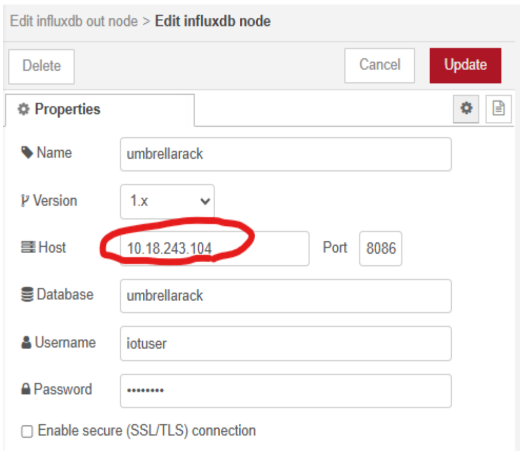
My PC IP: 10.19.8.210.

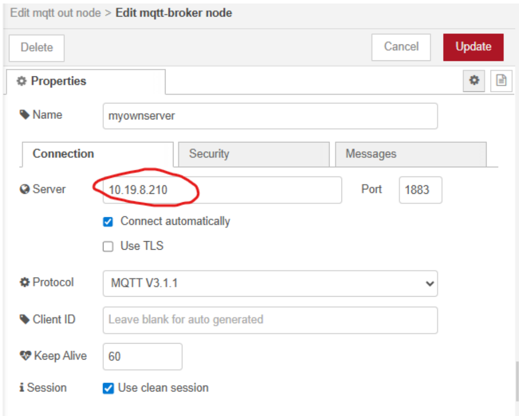
My Raspberry Pi IP: 10.18.243.104.

1. Import node-red flow
2. Connect the hardware
   - Slot_1:
     1. Ultrasonic sensors ➔ Digital port 8
     2. Green LED ➔ Digital port 6
     3. Red LED ➔ Digital port 4
   - Slot_2:
     1. Ultrasonic sensors ➔ Digital port 7
     2. Green LED ➔ Digital port 5
     3. Red LED ➔ Digital port 3

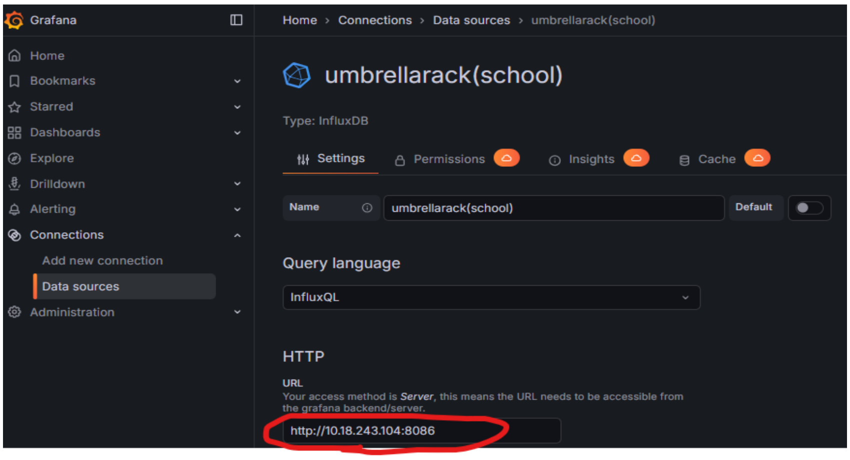
   Buzzer ➔ Digital port 2
3. Change the server IP to your own raspberry pi IP. You have to do this every time when you use a different IP.

   Update the MQTT and DB node configurations. Note that the IPs will differ because the MQTT broker is on the PC, while InfluxDB is on the Raspberry Pi.





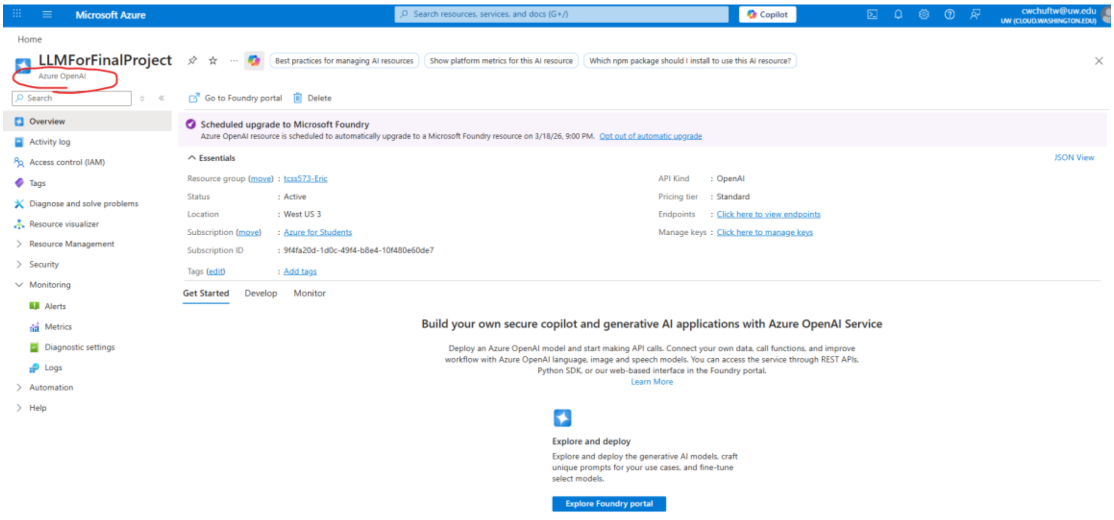
Change Grafana ➔ Connections ➔ Data sources ➔ DB ➔ URL.



Change the IP in python file(smartUmbrellaRackEn.py).

```python
# === InfluxDB 1.x Settings ===
INFLUX_HOST = "10.18.243.104" # This is your raspberry Pi's IP address.
INFLUX_PORT = 8086
INFLUX_DB = "umbrellarack"
```

4. Put your API KEY in “send Picture To LLM” node & “set zip code” node.
5. Create Azure OpenAI service



6. Put your Azure OpenAI API KEY in “sendPictureToLLM” node
7. Run the MQTT Broker and subscribe specific channel on your PC (Fog Layer).

Windows PowerShell ➔ cd "C:\Program Files\Mosquitto"
➔ .\mosquitto_sub.exe -h localhost -t "umbrella/slot1/photo" -v

You can run this on your raspberry pi to test if the connection is success. (pi@iotkit-EricC:~ $ mosquitto_pub -h 10.19.8.210(Your Broker IP) -t "umbrella/slot1/photo" -m "Hello from Raspberry Pi!")

If the connect success you should see the message on your windows powershell.

```
PS C:\Program Files\Mosquitto> .\mosquitto_sub.exe -h localhost -t "umbrella/slot1/photo" -v
umbrella/slot1/photo Hello from Raspberry Pi!
umbrella/slot1/photo Hello from Raspberry Pi!
```

8. Using python virtual environment to run the python file.
   Activate Venv & Run smartUmbrellaRackEn.py
   (.\venv\Scripts\Activate.ps1)
   (python .\smartUmbrellaRackEn.py)

```
PS C:\Users\kenny3016eric\Desktop\Python\IOT_Final_Project> .\venv\Scripts\Activate.ps1
(venv) PS C:\Users\kenny3016eric\Desktop\Python\IOT_Final_Project> python .\smartUmbrellaRackEn.py
2026-03-09 17:29:58.796133: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see sligh
tly different numerical results due to floating-point round-off errors from different computation orders. To turn
them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2026-03-09 17:30:00.120942: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see sligh
tly different numerical results due to floating-point round-off errors from different computation orders. To turn
them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
WARNING:tensorflow:From C:\Users\kenny3016eric\Desktop\Python\IOT_Final_Project\venv\Lib\site-packages\tf_keras\sr
c\losses.py:2976: The name tf.losses.sparse_softmax_cross_entropy is deprecated. Please use tf.compat.v1.losses.sp
arse_softmax_cross_entropy instead.


[System] Restoring previous memory from database...
🔄 Restore successful: slot1 is currently empty.
🔄 Restore successful: slot2 is currently occupied.
----------------------------------------
🚀 Fog Node zero-registration anti-theft brain started, waiting for images...
✅ Successfully connected to MQTT Broker
```

Now the theft detection system is running. It will capture the image when someone put in or take away an umbrella from the umbrella rack. If the one who take away the umbrella is not the same person as the one who put in, the alarm system will be triggered.

Dashboards:

Dashboard 1.
Slot Status & Data Analysis

Dashboard 2.
LLM explain the picture

Dashboard 3.
Cloud Service Usage